

2(a)(i)	water shown formed from –OH and –H ; A H₂O formed without indicating where from peptide bond shown correctly between C of carboxylic acid and N of amino group ; complete dipeptide drawn ; A hydrogen 'up or down'	3
2(a)(ii)	R group / side chain / variable group / residual group / functional group ; indicate, different amino acids / type of amino acid ; A specific to each amino acid A the two amino acids are different A two examples of R groups, e.g. –H and –CH₃ both amino acids have different R groups = 2 marks	2
2(b)	<i>any two from</i> straight chain / linear, v helix / helical ; I coil(ed) R branched / branching β-glucose, not α-glucose ; A β-1:4 glycosidic bond v α-1:4 glycosidic bond (β-)glucose / monomers / residues, are arranged, rotated 180° to each other / AW ; AVP ; e.g. more hydrogen bonds	2

Question	Answer	Marks
2(c)	<i>any four from</i> 1 (molecules) form fibrils and fibres ; 2 hydrogen bonding between (cellulose) <u>molecules</u> ; 3 (cellulose molecule) is straight / linear ; 4 (straight chain allows) molecules lie parallel to each other ; <i>mp4 dependent on mp3</i> 5 gives strength (to cell wall) to, prevent cell bursting / withstand (turgor) pressure / AW ; R if only in context of a cellulose molecule 6 <i>ref.</i> to fibres at angles / criss-cross / AW ; 7 <i>idea of</i> many gaps, in wall / between fibres, allowing passage of water / (named) substances / making cell wall permeable ; I plasmodesmata R partially / AW, permeable 8 AVP ; e.g. cellulose is insoluble many –OH groups (for hydrogen bonding)	4